

**A Lattice Framework for Option Pricing with Two State Variables**

Phelim P. Boyle\*

**Abstract**

A procedure is developed for the valuation of options when there are two underlying state variables. The approach involves an extension of the lattice binomial approach developed by Cox, Ross, and Rubinstein to value options on a single asset. Details are given on how the jump probabilities and jump amplitudes may be obtained when there are two state variables. This procedure can be used to price any contingent claim whose payoff is a piece-wise linear function of two underlying state variables, provided these two variables have a bivariate lognormal distribution. The accuracy of the method is illustrated by valuing options on the maximum and minimum of two assets and comparing the results for cases in which an exact solution has been obtained for European options. One advantage of the lattice approach is that it handles the early exercise feature of American options. In addition, it should be possible to use this approach to value a number of financial instruments that have been created in recent years.

**I. Introduction**

The option pricing framework has proved extremely useful for the analysis of many aspects of corporate finance and investment practice. The volume and variety of options traded on world exchanges has expanded enormously since Black and Scholes developed their seminal approach. Corporations in many countries now offer increasingly sophisticated instruments to the investing public. Budd (1983) describes a number of these new types of securities. The valuation of these securities poses challenging problems since they may involve a complex package of imbedded options whose payoffs depend on several state variables.

Options involving two state variables have been discussed by Stulz (1982), Johnson (1981), Schwartz (1982), and Boyle and Kirzner (1985). These papers concentrated on the valuation of European options and were able to develop ana-

\* School of Accounting, University of Waterloo, Waterloo, Ontario, Canada N2L 3G1. The author is grateful to Don Haley for superb research assistance and to the Natural Sciences and Engineering Research Council of Canada, the Centre for Accounting Research and Education at the University of Waterloo, and the Financial Research Foundation of Canada for support. He is also grateful to John Hull, Gordon Willmot, and Alan White for helpful discussions and to Jeremy Envine and Mark Rubinstein for useful comments. An earlier version of this paper was presented at the 1986 European Finance Association Meeting in Dublin and the author thanks Marshall Blume for constructive comments. The author also acknowledges helpful suggestions from an anonymous *JFQA* referee.

lytic or quasi-analytic solutions involving the bivariate normal density function. American options with payoffs depending on two state variables satisfy the same basic generalization of the Black-Scholes equation as European options. However, the boundary conditions are more complex in the case of American options to accommodate the possibility of early exercise. It is possible to solve the equation for American options in the case of more than one state variable using a finite difference approach. With two or more state variables, the computations quickly become quite expensive in the case of the finite difference method. In addition, there are significant start-up tests to develop an efficient computing algorithm.

In the case of the valuation of American options when there is only one state variable, the lattice binomial approach developed by Cox, Ross, and Rubinstein (1979) has a strong intuitive appeal. It is extremely simple to implement. The basic idea is to replace the continuous distribution of stock prices by a two-point discrete distribution over successively smaller time intervals. Convergence to the true option value is obtained by increasing the number of steps. The terminal distribution of the stock price is developed using a binomial lattice. This method has won widespread acceptance in the case of a single state variable. Geske and Shastri (1985) outline its advantages and drawbacks as well as those of a number of other approximation techniques in the case of options with a single state variable.

The basic purpose of the present paper is to develop an extension of the Cox, Ross, Rubinstein (CRR) lattice binomial algorithm to handle the situation in which the payoff from the option depends on more than one state variable. Specifically, we consider the case in which the option is a function of two underlying state variables although it is possible that the procedure can be extended to situations involving a higher number of state variables. As a prelude to the development of a lattice approach in the case of two state variables, it proved convenient to modify the CRR approach in the case of one state variable. This modification consisted of replacing the two-point jump process used by CRR with a three-jump process. When this was done, we obtained a more efficient algorithm for the valuation of options when there is just one state variable; a result that may be of some interest in its own right.

Section II of the paper describes the modification of the CRR approach in the case of a single state variable. Section III develops a lattice type approach for the valuation of options whose payoff depends on two state variables. It turns out that a five-point jump process enables us to develop an appropriate lattice in three dimensions (two asset dimensions and one time dimension). We discuss the determination of the jump probabilities and jump amplitudes and are able to draw on some of the results of Section II in the development.

Numerical results are given in Section IV. These include cases in which exact solutions (available from Stulz (1982)), are used to illustrate the accuracy of the procedure. We also derive values for American put options on the minimum of two assets and illustrate the value of the early exercise provision.

The final section contains some concluding comments.

## II. Modification of the Two State Approach for a Single State Variable

In this section, we develop a modification of the CRR lattice binomial approach for option valuation in the case of a single state variable. The modification introduced will be helpful in generalizing the approach to two state variables. One of the features of the CRR approach is that by assuming that the stock price can move only either upward or downward, the relationship between the hedging argument and the mathematical development is especially clear. Their approach provides a powerful demonstration that options can be priced by discounting their expected value in a risk-neutral world. However, if we know the distributional assumptions and are assured that a risk-neutral valuation procedure is appropriate, then other types of discrete approximation can be used. We can regard the option valuation problem as a problem in numerical analysis and replace the continuous distribution of stock prices by a suitable discrete process as long as the discrete distribution tends to the appropriate limit.<sup>1</sup> In this section, we use a three-jump process instead of the two-jump process used in the CRR method.

Three-jump models have been used before in the literature to analyze option valuation problems. Stapleton and Subrahmanyam (1984) discussed a three-jump model, but they did not examine its numerical efficiency or solve for the jump probabilities. Parkinson (1977) employed a three-jump model to value the American put, but his approach seems difficult to generalize to situations involving more than one state variable. Brennan and Schwartz (1978) related the coefficients of a transformed Black-Scholes equation to the probabilities of a three-jump process.

To proceed with our development, let us consider an asset ( $S$ ) with a lognormal distribution of returns. Over a small time interval this distribution is approximated by a three-point jump process in such a way that the expected return on the asset is the riskless rate, and the variance of the discrete distribution is equal to the variance of the corresponding lognormal distribution. It is convenient to employ the following notation:

- $T$  = time to option maturity (in years),
- $X$  = exercise price of option,
- $r$  = the continuously compounded yearly interest rate,
- $\sigma^2$  = the variance of the rate of return on the underlying asset (yearly),
- $n$  = the number of time steps into which the interval of length  $T$  is divided,
- $h = T/n$ : length of one time step,
- $S_u$  = asset value after an up jump,
- $S$  = asset value after a horizontal jump,
- $S_d$  = asset value after a down jump.

---

<sup>1</sup> Proofs that the approximating discrete distribution tends to the appropriate continuous distribution for both the one-state-variable case and the two-state-variable case are available from the author. The author thanks the referee for affirming the importance of this convergence.

For the three-jump process considered in this section, the continuous-return distribution is approximated by the following discrete distribution:

| <u>Nature of Jump</u> | <u>Probability</u> | <u>Asset Price</u> |
|-----------------------|--------------------|--------------------|
| Up                    | $p_1$              | $Su$               |
| Horizontal            | $p_2$              | $S$                |
| Down                  | $p_3$              | $Sd$               |

It is convenient to impose the condition that  $ud = 1$ . Our task is to find suitable values for the probabilities  $p_1$ ,  $p_2$ , and  $p_3$  and the stretch parameter  $u$  in terms of the known variables. This can be achieved by first imposing three conditions to solve for the probabilities in terms of  $u$  and the other variables. Then we show how a suitable value of  $u$  can be obtained.

The three conditions are:

- (i) the probabilities sum to one,
- (ii) the mean of the discrete distribution :  $SM$ , is equal to the mean of the continuous lognormal distribution

$$S \exp(rh) = SM, \text{ and}$$

- (iii) the variance of the discrete distribution :  $S^2V$ , is equal to the variance of the continuous distribution

$$S^2M^2 [\exp(\sigma^2 h) - 1] = S^2V.$$

Later on, when we specify the value of the stretch parameter  $u$ , we insist that each of the probabilities must be positive. The above three conditions are,

$$(1) \quad p_1 + p_2 + p_3 = 1,$$

$$(2) \quad p_1 Su + p_2 S + p_3 \frac{S}{u} = SM,$$

$$(3) \quad p_1 (S^2 u^2 - S^2 M^2) + p_2 (S^2 - S^2 M^2) + p_3 \left( \frac{S^2}{u^2} - S^2 M^2 \right) = S^2 V.$$

It is convenient to divide the second equation by  $S$  and the third by  $S^2$ . Then the first equation can be used to remove  $p_2$  from the last two equations. This yields

$$(4) \quad p_1 (u - 1) + p_3 \left( \frac{1}{u} - 1 \right) = M - 1,$$

$$(5) \quad p_1 (u^2 - 1) + p_3 \left( \frac{1}{u^2} - 1 \right) = V + M^2 - 1.$$

---

These equations can be solved to give explicit expressions for  $p_1$  and  $p_2$  as follows,

$$(6) \quad p_1 = \frac{(V + M^2 - M)u - (M - 1)}{(u - 1)(u^2 - 1)},$$

$$(7) \quad p_3 = \frac{u^2(V + M^2 - M) - u^3(M - 1)}{(u - 1)(u^2 - 1)},$$

$$(8) \quad p_2 = 1 - p_1 - p_3.$$

It is convenient to use the following notation,

$$(9) \quad p_1 = f(u, M, V),$$

$$(10) \quad p_3 = g(u, M, V).$$

Note that these equations imply that  $u$  is not equal to 1. This is reasonable since  $u$  corresponds to an up jump, and it will exceed 1.

So far,  $u$  has not been specified. In general, there will be a range of values of  $u$  that produce reasonable values for the probabilities. Recall that the CRR paper used

$$(11) \quad u = \exp[\sigma\sqrt{h}].$$

If we use this value of  $u$ , then for many realistic parameter values  $p_2$  will turn out to be negative. For example, suppose  $\sigma = 0.20$ ,  $r = 0.1$ ,  $T = 1.0$ , and  $n = 20$ . With these parameter values, Equation (11) implies that  $u = 1.045736$ , and the probabilities given by (6), (7), and (8) become  $p_1 = 0.553859$ ,  $p_2 = -0.018440$ , and  $p_3 = 0.464581$ . The value of  $u$  given by Equation (11) is too low to produce a positive value for  $p_2$  given the parameters of this example.<sup>2</sup> To rectify this, define  $u$  as

$$(12) \quad u = \exp[\lambda\sigma\sqrt{h}],$$

where  $\lambda$  is greater than 1. By using different values for  $\lambda$ , a range of values of  $u$  is obtained, and there is an interval within this range that produces acceptable values for all the probabilities and  $p_2$  in particular. Table 1 displays the values of  $u$  and the corresponding probabilities for a range of values of  $\lambda$ .

In order to determine the most effective value of  $\lambda$  to use, a number of options were evaluated over a range of parameter values. Best results were obtained when the probabilities were roughly equal. Although the primary aim in

<sup>2</sup> Recall that for the CRR two-jump approximation, the variance of the discrete distribution is biased, whereas this is not the case for the three-jump process considered here. The stretch factor " $u$ " will not be the same for both approximations to give meaningful results.

TABLE 1  
Jump Amplitudes and Jump Probabilities  
 $\sigma = 0.2, r = 0.1, T = 1.0, n = 20$

| $\lambda$ | $u$    | $p_1$  | $p_3$  | $p_2$   |
|-----------|--------|--------|--------|---------|
| 1.00      | 1.0457 | 0.5539 | 0.4646 | -0.0184 |
| 1.10      | 1.0504 | 0.4610 | 0.3798 | 0.1592  |
| 1.20      | 1.0551 | 0.3900 | 0.3156 | 0.2943  |
| 1.30      | 1.0599 | 0.3346 | 0.2659 | 0.3995  |
| 1.40      | 1.0646 | 0.2904 | 0.2266 | 0.4829  |
| 1.50      | 1.0694 | 0.2547 | 0.1951 | 0.5502  |
| 1.60      | 1.0742 | 0.2253 | 0.1694 | 0.6053  |
| 1.70      | 1.0790 | 0.2008 | 0.1482 | 0.6510  |
| 1.80      | 1.0838 | 0.1802 | 0.1305 | 0.6892  |
| 1.90      | 1.0887 | 0.1627 | 0.1156 | 0.7216  |
| 2.00      | 1.0936 | 0.1477 | 0.1030 | 0.7493  |

developing the three-jump lattice approach to option valuation in the case of one state variable was to pave the way for extensions to two or more underlying assets, it may be of interest to summarize the results obtained by this algorithm with those of the two-jump approach used by CRR. For a range of parameter values, we found that the accuracy of the three-jump method with 5 time intervals was comparable to that of the CRR method with 20 time intervals.<sup>3</sup>

### III. A Lattice Model for Valuation of Options on Two Underlying Assets

In this section, we develop a valuation algorithm for the pricing of options when there are two underlying assets. The method used to compute the jump probabilities within the lattice is outlined, and a procedure for selecting the jump amplitudes is explained. The essence of the idea is to derive a generalization of the method developed in the previous section.

It is assumed that the joint density of the two underlying assets is a bivariate lognormal distribution. When we implement the risk-neutral valuation procedure, it is implied that both assets earn the riskless rate. To specify the distribution, we need the variance-covariance matrix as well. The approximating discrete distribution has the same expected values and variance-covariance values as the discrete distribution. This means that one needs at least five degrees of freedom in constructing the discrete approximating distribution. In addition, we need to ensure that the jump probabilities add up to one, which gives another condition to be satisfied. A five-point jump process can be constructed to satisfy the various requirements and can be used to generate a two  $\times$  one-dimensional lattice suitable for valuing the options we are considering.

It is convenient to rely on the basic notation used in the previous section. We use the subscript 1 to denote the first asset and the subscript 2 to denote the second asset:

<sup>3</sup> For space reasons, summary comparison tables are not included in this paper, but are available from the author on request.

|               | Asset 1      | Asset 2      |
|---------------|--------------|--------------|
| Current value | $S_1$        | $S_2$        |
| Variance rate | $\sigma_1^2$ | $\sigma_2^2$ |

We assume that the option matures after time  $T$  and that its exercise price is  $X$ . We also assume that the time interval  $[0, T]$  is divided into  $n$  equal steps each of length  $h$ . As before, the continuously compounded riskless rate of interest is  $r$  per annum. Suppose we consider a time interval of length  $h$ . The means and variances of the two assets at the end of this interval are obtained from the properties of the lognormal distribution and are:

|          | Asset 1     | Asset 2     |
|----------|-------------|-------------|
| Mean     | $S_1 M_1$   | $S_2 M_2$   |
| Variance | $S_1^2 V_1$ | $S_2^2 V_2$ |

where

$$(13) \quad M_i = \exp[rh] ,$$

$$(14) \quad V_i = M_i^2 [\exp(\sigma_i^2 h) - 1] ,$$

$$i = 1, 2 .$$

The expected value of the product of the two variables  $S_1$  and  $S_2$  can be written down using the properties of the joint lognormal distribution

$$(15) \quad E(S_1 S_2) = S_1 S_2 M_1 M_2 [\exp(\rho \sigma_1 \sigma_2 h)]$$

$$= S_1 S_2 R .$$

From this last equation, we can derive the covariance of the two assets. Thus, we have the means and variance-covariance matrix of the bivariate lognormal distribution of asset values at the end of the time interval  $h$ . To replace this distribution with a discrete distribution, we ensure that the first and second moments (including the covariance term) of the approximating discrete distribution are equal to those of the continuous distribution.

In order to obtain an efficient algorithm, we found that a five-point jump process was the most suitable. Given that the pair of assets has current value  $(S_1, S_2)$ , there are five distinct outcomes for the proposed jump process. We can summarize the outcomes of the jump process as follows:

| Event | Probability of Event | Value of Asset given that Event has Occurred |           |
|-------|----------------------|----------------------------------------------|-----------|
|       |                      | Asset 1                                      | Asset 2   |
| $E_1$ | $p_1$                | $S_1 u_1$                                    | $S_2 u_2$ |
| $E_2$ | $p_2$                | $S_1 u_1$                                    | $S_2 d_2$ |
| $E_3$ | $p_3$                | $S_1 d_1$                                    | $S_2 d_2$ |
| $E_4$ | $p_4$                | $S_1 d_1$                                    | $S_2 u_2$ |
| $E_5$ | $p_5$                | $S_1$                                        | $S_2$     |

Thus, in analogy to the earlier case,  $u_1$  represents the stretch factor for asset one, and  $u_2$  represents the stretch factor for asset two. We impose the conditions

$$u_1 d_1 = 1 , \quad u_2 d_2 = 1 .$$

Assume the process starts when the asset values are  $S_1$  and  $S_2$ . We envisage a three-dimensional representation with time along the vertical axis. The asset price pair can take one of five values after time  $h$  depending on which of  $E_1, E_2, E_3, E_4$ , or  $E_5$  has occurred. These five points correspond to points in  $S_1$ - $S_2$  space lying in a plane directly above the point of origin. The first four points lie at the corners of the rectangle and the final point lies in the center of the rectangle vertically above the point of origin. As this process is repeated, we generate a three-dimensional lattice structure resembling an inverted pyramid. The terminal distribution of the pair of assets is represented by the topmost of a layered series of rectangles in which the center of each rectangle lies vertically above the point of origin.

To evaluate the probabilities and the stretch parameters, it is convenient to proceed as we did in the previous section: first, find the probabilities in terms of  $u_1, u_2$ , and the other variables. By equating the mean of the discrete distribution to the mean of the continuous distribution, we obtain two equations

$$(16) \quad (p_1 + p_2)S_1 u_1 + p_5 S_1 + (p_3 + p_4)S_1 d_1 = S_1 M_1 ,$$

$$(17) \quad (p_1 + p_4)S_2 u_2 + p_5 S_2 + (p_2 + p_3)S_2 d_2 = S_2 M_2 .$$

Two more equations result from equating the variances

$$(18) \quad (p_1 + p_2)(S_1^2 u_1^2 - S_1^2 M_1^2) + p_5(S_1^2 - S_1^2 M_1^2) \\ + (p_3 + p_4)(S_1^2 d_1^2 - S_1^2 M_1^2) = S_1^2 V_1 ,$$

$$(19) \quad (p_1 + p_4)(S_2^2 u_2^2 - S_2^2 M_2^2) + p_5(S_2^2 - S_2^2 M_2^2) \\ + (p_2 + p_3)(S_2^2 d_2^2 - S_2^2 M_2^2) = S_2^2 V_2 .$$

Since the probabilities sum to 1 we also have

$$(20) \quad p_1 + p_2 + p_3 + p_4 + p_5 = 1 .$$

Now we can use the results of the previous section to simplify the analysis. Note that there is a strong correspondence between the three equations (1), (2), and (3) and the triplet (20), (16), and (18) and also with the triplet (20), (17), and (19). In fact, if we regard  $(p_1 + p_2)$ ,  $(p_3 + p_4)$ , and  $p_5$  as new probabilities, we can solve for these new probabilities. The explicit expressions are given by Equations (6), (7), and (8) with appropriate modifications. In the same way, we can solve Equations (20), (17), and (19) for the "new" probabilities  $(p_1 + p_4)$ ,  $(p_2 + p_3)$ , and  $p_5$ . Notice that this solution procedure corresponds to the projec-

tion of the two-variate distribution onto the unconditional distribution of a single variate. Using this procedure, we obtain the following results

$$(21) \quad p_1 + p_2 = f(u_1, M_1, V_1) = f_1 ,$$

$$(22) \quad p_3 + p_4 = g(u_1, M_1, V_1) = g_1 ,$$

$$(23) \quad p_1 + p_4 = f(u_2, M_2, V_2) = f_2 ,$$

$$(24) \quad p_2 + p_3 = g(u_2, M_2, V_2) = g_2 .$$

In view of Equation (20), we have two distinct expressions for  $p_5$  so that the following consistency equation needs to be satisfied

$$(25) \quad f_1 + g_1 = f_2 + g_2 .$$

This gives a relationship between  $u_1$  and  $u_2$  that must be satisfied.

The final equation is obtained from relating the expected value of the product of the two assets under the continuous distribution and the approximating discrete distribution. Using Equation (15), this gives

$$(26) \quad (p_1 u_1 u_2 + p_2 u_1 d_2 + p_3 d_1 d_2 + p_4 d_1 u_2 + p_5) S_1 S_2 = R S_1 S_2 .$$

Dividing across by  $S_1 S_2$  and eliminating  $p_5$  yields

$$p_1(u_1 u_2 - 1) + p_2(u_1 d_2 - 1) + p_3(d_1 d_2 - 1) + p_4(d_1 u_2 - 1) = R - 1 .$$

If we substitute for  $d_1$  and  $d_2$  and use Equations (21), (22), (23), and (24), we can use this last equation to obtain the following expression for  $p_1$

$$(27) \quad p_1 = \frac{u_1 u_2 (R - 1) - f_1(u_1^2 - 1) - f_2(u_2^2 - 1) + (f_2 + g_2)(u_1 u_2 - 1)}{(u_1^2 - 1)(u_2^2 - 1)} .$$

Armed with this expression for  $p_1$ , we can find  $p_2$ ,  $p_3$ , and  $p_4$  using Equations (21), (22), (23), and (24). The results are

$$(28) \quad p_2 = \frac{f_1(u_1^2 - 1)u_2^2 + f_2(u_2^2 - 1) - (f_2 + g_2)(u_1 u_2 - 1) - u_1 u_2 (R - 1)}{(u_1^2 - 1)(u_2^2 - 1)} ,$$

$$(29) \quad p_3 = \frac{u_1 u_2 (R - 1) - f_1(u_1^2 - 1)u_2^2 + g_2(u_2^2 - 1)u_1^2 + (f_2 + g_2)(u_1 u_2 - u_2^2)}{(u_1^2 - 1)(u_2^2 - 1)} ,$$

$$(30) \quad p_4 = \frac{f_1(u_1^2 - 1) + f_2(u_2^2 - 1)u_1^2 - (f_2 + g_2)(u_1 u_2 - 1) - u_1 u_2 (R - 1)}{(u_1^2 - 1)(u_2^2 - 1)} .$$

To illustrate the computation of these probabilities in a specific case, let us assume the following parameters:  $\sigma_1 = 0.2$ ,  $\sigma_2 = 0.25$ ,  $r = 0.1$ ,  $n = 20$ ,  $T = 1.0$ , and  $\rho = 0.5$ . Let

$$u_1 = \exp[(1.1)\sigma_1\sqrt{h}]$$

so that for these parameter values

$$u_1 = 1.05042358 .$$

In order to ensure consistency, Equation (25) must be satisfied. Equation (25) gives a relationship between  $u_1$  and  $u_2$ . In order to solve for  $u_2$ , we use Newton's method and take the initial value of  $u_2$  to be:

$$\begin{aligned} u_2 &= \exp[(1.1)\sigma_2\sqrt{h}] \\ &= 1.06342185. \end{aligned}$$

It turns out that the value of  $u_2$  that satisfies Equation (25) is close to this starting value. When  $u_2 = 1.0632918$ , Equation (25) is satisfied. Armed with the values of  $u_1$  and  $u_2$ , we can proceed to compute the probabilities. These are given for a range of values of  $\lambda$  in Table 2. Probabilities for the particular  $(u_1, u_2)$  combination just discussed are displayed in the second line of Table 2. Note from Table 2 that the sum of the probabilities  $p_1$  and  $p_2$  corresponds to the probability of an upward jump in the case of the single asset given by  $p_1$  in Table 1. There is a similar relationship between the sum  $(p_3 + p_4)$  from Table 2 and the probability of a down jump in Table 1. This illustrates the relationship between the bivariate distribution for  $S_1$  and  $S_2$  and the univariate unconditional distribution for  $S_1$  and the univariate unconditional distribution for  $S_2$ .

TABLE 2  
Jump Amplitudes and Jump Probabilities for 5-Jump Process  
 $\sigma_1 = 0.2, \sigma_2 = 0.25, r = 0.1, n = 20, T = 1, \rho = 0.5$

| $\lambda$ | $u_1$  | $u_2$  | $p_1$  | $p_2$  | $p_3$  | $p_4$  | $p_5$   |
|-----------|--------|--------|--------|--------|--------|--------|---------|
| 1.00      | 1.0457 | 1.0574 | 0.4201 | 0.1337 | 0.3448 | 0.1198 | -0.0184 |
| 1.10      | 1.0504 | 1.0633 | 0.3499 | 0.1111 | 0.2814 | 0.0984 | 0.1592  |
| 1.20      | 1.0551 | 1.0692 | 0.2962 | 0.0938 | 0.2334 | 0.0822 | 0.2943  |
| 1.30      | 1.0599 | 1.0752 | 0.2543 | 0.0803 | 0.1963 | 0.0696 | 0.3995  |
| 1.40      | 1.0646 | 1.0812 | 0.2208 | 0.0696 | 0.1670 | 0.0597 | 0.4829  |
| 1.50      | 1.0694 | 1.0873 | 0.1937 | 0.0609 | 0.1434 | 0.0517 | 0.5502  |
| 1.60      | 1.0742 | 1.0934 | 0.1715 | 0.0538 | 0.1243 | 0.0451 | 0.6053  |
| 1.70      | 1.0790 | 1.0995 | 0.1529 | 0.0479 | 0.1085 | 0.0397 | 0.6510  |
| 1.80      | 1.0838 | 1.1056 | 0.1373 | 0.0429 | 0.0953 | 0.0352 | 0.6892  |
| 1.90      | 1.0887 | 1.1118 | 0.1240 | 0.0387 | 0.0842 | 0.0314 | 0.7216  |
| 2.00      | 1.0936 | 1.1180 | 0.1126 | 0.0351 | 0.0748 | 0.0282 | 0.7493  |

#### IV. Numerical Examples

In this section, we present numerical examples to illustrate the operation of the method developed in the previous section. Both Johnson (1981) and Stulz (1982) have derived a number of exact results for European options when there

are two underlying assets. These results can be used to check the accuracy of the approach. In particular, expressions exist for a European call option on the maximum of two assets and for a European put option on the minimum of two assets. Johnson (1985) has obtained a generalization of these results for the case involving a European call option on the maximum or minimum of several assets. His results could be used as benchmarks to check extensions of our method to situations involving three or more assets.

The advantage of the present approach is that it permits early exercise and thus can be used to value American options and, in particular, American put options. Our approach can also be modified to handle dividends and other pay-outs.

For our numerical solutions, we assume  $S_1 = 40$ ,  $S_2 = 40$ ,  $\sigma_1 = 0.20$ ,  $\sigma_2 = 0.30$ ,  $\rho = 0.5$ ,  $r = 5$ -percent per annum effective  $= 0.048790$  continuously,  $T = 7$  months  $= 0.583333$  years, and exercise prices  $= 35, 40, 45$ . We display in Table 3 the results for three types of options: European call options on the maximum of the two assets; European put options on the minimum of the two assets; and American put options on the minimum of two assets. Since we are assuming no dividend payments, the value of the European call on the maximum of the two assets will be (identical to) the value of an American call option with the same specifications. The agreement between the numbers obtained using the lattice approximation and the accurate values is quite good, especially when 50 time steps are used. In this case, the maximum difference is 0.005 and the accuracy would be adequate for most applications. For this set of parameter values, the American put is not much more valuable than its European counterpart.

TABLE 3  
Different Types of Options Involving Two Assets:  
Comparison of Lattice Approach of this Paper with Accurate Results

|                       | <u>Number of Steps</u>                                |           |           |                       |
|-----------------------|-------------------------------------------------------|-----------|-----------|-----------------------|
| <u>Exercise Price</u> | <u>10</u>                                             | <u>20</u> | <u>50</u> | <u>Accurate Value</u> |
|                       | <i>European Call on the Maximum of the Two Assets</i> |           |           |                       |
| 35                    | 9.404                                                 | 9.414     | 9.419     | 9.420                 |
| 40                    | 5.466                                                 | 5.477     | 5.483     | 5.488                 |
| 45                    | 2.817                                                 | 2.790     | 2.792     | 2.795                 |
|                       | <i>European Put on the Minimum of the Two Assets</i>  |           |           |                       |
| 35                    | 1.425                                                 | 1.394     | 1.392     | 1.387                 |
| 40                    | 3.778                                                 | 3.790     | 3.795     | 3.798                 |
| 45                    | 7.475                                                 | 7.493     | 7.499     | 7.500                 |
|                       | <i>American Put on the Minimum of the Two Assets</i>  |           |           |                       |
| 35                    | 1.450                                                 | 1.423     | 1.423     |                       |
| 40                    | 3.870                                                 | 3.885     | 3.892     |                       |
| 45                    | 7.645                                                 | 7.674     | 7.689     |                       |

## V. Concluding Comments

We have seen how the lattice approach to option valuation that has been developed for options on a single state variable can be extended to handle situa-

tions in which there are two state variables. This means that American options in which there is a possibility of early exercise can be handled using this technique. It is suggested that approximation procedures of this type may be useful in the valuation of complex securities involving options that depend on two state variables. Thus, for example, the procedure developed here could be used in the pricing of secured debt when the underlying firm consists of two risky assets with a joint lognormal distribution. Recently, Stulz and Johnson (1985) have examined this problem and used a numerical approach to solve the resulting partial differential equation. Another area in which this new valuation technique might be useful is in the valuation of options with stochastic volatilities because, in this case, there are two stochastic state variables. Recently, a number of papers have appeared on the valuation of such options (c.f., Wiggins (1985), Hull and White (1985), Johnson and Shanno (1985)). To date, the techniques used to derive option values have involved the direct solution of the partial differential equation (Wiggins (1985)) or Monte-Carlo approaches (Johnson and Shanno (1985) and Hull and White (1985)). The method suggested in this paper may provide another approach.

## References

- Boyle, P. P., and E. F. Kirzner. "Pricing Complex Options: Echo-Bay Ltd. Gold Purchase Warrants." *Canadian Journal of Administrative Sciences*, 2 (Dec. 1985), 294–306.
- Brennan, M. J., and E. S. Schwartz. "Finite Difference Methods and Jump Processes Arising in the Pricing of Contingent Claims: A Synthesis." *Journal of Financial and Quantitative Analysis*, 13 (Sept. 1978), 461–474.
- Budd, N. "The Future of Commodity Indexed Financing." *Harvard Business Review*, 61 (July–Aug. 1983), 44–50.
- Cox, J.; S. Ross; and M. Rubinstein. "Option Pricing: A Simplified Approach." *Journal of Financial Economics*, 7 (Oct. 1979), 229–264.
- Geske, R., and H. E. Johnson. "The American Put Option Valued Analytically." *The Journal of Finance*, 39 (Dec. 1984), 1511–1524.
- Geske, R., and K. Shastri. "Valuation by Approximation: A Comparison of Alternative Option Valuation Techniques." *Journal of Financial and Quantitative Analysis*, 20 (March 1985), 45–71.
- Hull, J., and A. White. "The Pricing of Options on Assets with Stochastic Volatilities." *Journal of Finance*, 42 (June 1987), 281–300.
- Johnson, H. "Pricing Complex Options." Mimeo, Dept. of Finance, College of Business Administration, Louisiana State Univ., Baton Rouge, LA (1981).
- . "Options on the Maximum or the Minimum of Several Assets." *Journal of Financial and Quantitative Analysis*, 22 (Sept. 1987), 277–283.
- Johnson, H., and D. Shanno. "Option Pricing when the Variance is Changing." *Journal of Financial and Quantitative Analysis*, 22 (June 1987), 143–151.
- Parkinson, M. "Option Pricing: The American Put." *Journal of Business*, 50 (Jan. 1977), 21–36.
- Schwartz, E. S. "The Pricing of Commodity Linked Bonds." *Journal of Finance*, 37 (May 1982), 525–541.
- Stapleton, R. C., and M. G. Subrahmanyam. "The Valuation of Options when Asset Returns are Generated by a Binomial Process." *The Journal of Finance*, 39 (Dec. 1984), 1525–1540.
- Stulz, R. M. "Options on the Minimum or the Maximum of Two Risky Assets: Analysis and Applications." *Journal of Financial Economics*, 10 (July 1982), 161–185.
- Stulz, R. M., and H. Johnson. "An Analysis of Secured Debt." *Journal of Financial Economics*, 14 (Dec. 1985), 501–521.
- Wiggins, J. "Stochastic Variance Option Pricing." Working Paper, Sloan School of Management, Massachusetts Institute of Technology, Cambridge, MA (1985).